Solving linear equations

Use inverse operations to solve each equation. Show your working in the space provided.

One-step equations

1. Solve each equation.

   (a) $x + 7 = 15$ (b) $y - 9 = 4$ (c) $5a = 35$

   (d) $\frac{b}{4} = 6$ (e) $m + 3 = -2$ (f) $-3n = 18$

2. For each of the following:

   (i) State the inverse operation needed.

   (ii) Solve the equation and check by substitution.

   (a) $p - 6 = 11$ (b) $\frac{q}{5} = 3$

Two-step equations

3. (a) $2x + 5 = 19$ (b) $3y - 4 = 20$ (c) $\frac{a}{4} + 3 = 8$

   (d) $5b + 2 = -13$ (e) $12 - 2m = 4$ (f) $\frac{n}{3} - 7 = -2$

4. Solve $7 - \frac{x}{2} = 1$. Check your solution by substituting it into the original equation. [3 marks]

Explain your reasoning

5. Read each part carefully.

(a) Alex says that the solution of $3x + 6 = 21$ is $x = 9$. Explain the error, then find the correct solution.

(b) Sam solves $\frac{x}{4} - 2 = 5$ by adding 2 to both sides first.

Explain why this is a useful first step, then finish the solution.

Equations with brackets and equations with pronumerals on both sides

6. Expand the brackets where needed, then collect like terms and solve.

(a) $3(x + 2) = 21$

(b) $2(y - 4) = 10$

(c) $5a + 4 = 2a + 19$

(d) $7b - 8 = 3b + 12$

(e) $2(m + 3) = m + 11$

(f) $3(n - 2) = 2n + 1$

7. Show a clear sequence of equivalent equations.

(a) $2(x + 5) = 26$ [1 mark]

(b) (i) Expand the brackets and solve $4(y - 1) = 2y + 10$. [2 marks]

(ii) Check your solution by substitution. [1 mark]

Equations from words

8. Three more than twice a number is 27. Let n represent the number. Write an equation and solve it to find the number. [3 marks]

9. Translate each statement into an equation using n for the number and solve the equation to find the number.

   (a) Five less than three times a number is 16.

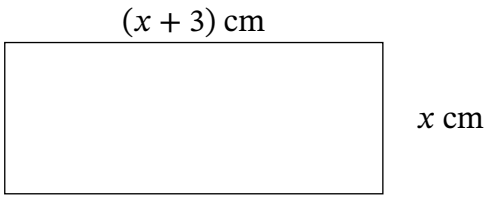
   (b) A number divided by four, then increased by six, gives 11.

Practical contexts

10. A bike hire shop charges a fixed fee of \$8 plus \$6 per hour. Mia pays \$38.

    Let h be the number of hours. Write and solve an equation to find how long Mia hired the bike. [4 marks]

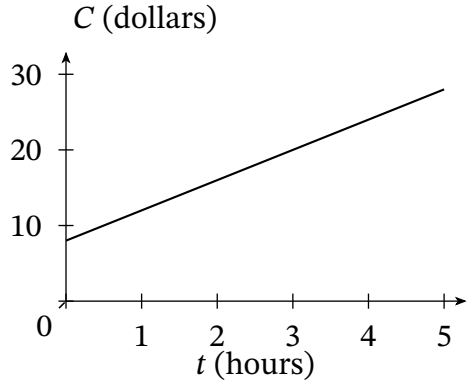
11. A rectangle has width x cm and length $(x + 3)$ cm. Its perimeter is 30 cm. Write an equation and find both dimensions.



12. A delivery service charges a fixed booking fee and \$2 per kilometre. Use the table to find the booking fee, then find the distance travelled when the total cost is \$23.

Distance (km)	2	4	6
Total cost (\$)	11	15	19

13. The graph shows the cost C dollars of hiring equipment for t hours. The relation is $C = 4t + 8$. Use an equation to find the time when the cost is \$24, then check it on the graph.



Answers

1. (a) $x = 8$ (b) $y = 13$ (c) $a = 7$
   (d) $b = 24$ (e) $m = -5$ (f) $n = -6$
2. (a) (i) Add 6.
   (ii) $p = 17$; $17 - 6 = 11$.
   (b) (i) Multiply by 5.
   (ii) $q = 15$; $\frac{15}{5} = 3$.
3. (a) $x = 7$ (b) $y = 8$ (c) $a = 20$
   (d) $b = -3$ (e) $m = 4$ (f) $n = 15$
4. $x = 12$.
5. (a) Subtract 6 before dividing by 3; $x = 5$.
   (b) Adding 2 undoes subtracting 2; $x = 28$.
6. (a) $x = 5$ (b) $y = 9$ (c) $a = 5$
   (d) $b = 5$ (e) $m = 5$ (f) $n = 7$
7. (a) $x = 8$
   (b) (i) $y = 7$
   (ii) Both sides equal 24.
8. $2n + 3 = 27$; $n = 12$.
9. (a) $3n - 5 = 16$; $n = 7$.
   (b) $\frac{n}{4} + 6 = 11$; $n = 20$.
10. $8 + 6h = 38$; $h = 5$. The hire lasted 5 hours.
11. $2x + 2(x + 3) = 30$; $x = 6$. Width 6 cm; length 9 cm.
12. Booking fee \$7; $7 + 2d = 23$; $d = 8$ km.
13. $4t + 8 = 24$; $t = 4$ hours.

Solutions

1. (a) $x + 7 = 15$
$x = 15 - 7$
$x = 8$

(b) $y - 9 = 4$
$y = 4 + 9$
$y = 13$

(c) $5a = 35$
$a = \frac{35}{5}$
$a = 7$

(d) $\frac{b}{4} = 6$
$b = 6 \times 4$
$b = 24$

(e) $m + 3 = -2$
$m = -2 - 3$
$m = -5$

(f) $-3n = 18$
$n = \frac{18}{-3}$
$n = -6$

2. (a) (i) Add 6 to both sides to undo subtracting 6.

(ii) $p - 6 = 11$
$p = 11 + 6$
$p = 17$
Check: $17 - 6 = 11$, which equals the right-hand side.

(b) (i) Multiply both sides by 5 to undo dividing by 5.

(ii) $\frac{q}{5} = 3$
$q = 3 \times 5$
$q = 15$
Check: $\frac{15}{5} = 3$, which equals the right-hand side.

3. (a) $2x + 5 = 19$
$2x = 14$
$x = 7$

(b) $3y - 4 = 20$
$3y = 24$
$y = 8$

(c) $\frac{a}{4} + 3 = 8$
$\frac{a}{4} = 5$
$a = 20$

(d) $5b + 2 = -13$
$5b = -15$
$b = -3$

(e) $12 - 2m = 4$
$-2m = -8$
$m = 4$

(f) $\frac{n}{3} - 7 = -2$
$\frac{n}{3} = 5$
$n = 15$

4. $7 - \frac{x}{2} = 1$
$-\frac{x}{2} = -6$
$x = 12$
Check: $7 - \frac{12}{2} = 7 - 6 = 1$, which equals the right-hand side.

5. (a) Substituting Alex's value gives $3 \times 9 + 6 = 33$, not 21, so $x = 9$ is incorrect. To isolate x, first subtract 6 from both sides, then divide both sides by 3.

$$\begin{aligned} 3x + 6 &= 21 \\ 3x &= 15 \\ x &= 5 \end{aligned}$$

(b) Adding 2 to both sides undoes subtracting 2 and leaves $\frac{x}{4}$ on its own. Then multiply both sides by 4.

$$\begin{aligned} \frac{x}{4} - 2 &= 5 \\ \frac{x}{4} &= 7 \\ x &= 28 \end{aligned}$$

6. (a)
$$\begin{aligned} 3(x + 2) &= 21 \\ 3x + 6 &= 21 \\ 3x &= 15 \\ x &= 5 \end{aligned}$$

(b)
$$\begin{aligned} 2(y - 4) &= 10 \\ 2y - 8 &= 10 \\ 2y &= 18 \\ y &= 9 \end{aligned}$$

(c)
$$\begin{aligned} 5a + 4 &= 2a + 19 \\ 3a + 4 &= 19 \\ 3a &= 15 \\ a &= 5 \end{aligned}$$

(d)
$$\begin{aligned} 7b - 8 &= 3b + 12 \\ 4b - 8 &= 12 \\ 4b &= 20 \\ b &= 5 \end{aligned}$$

(e)
$$\begin{aligned} 2(m + 3) &= m + 11 \\ 2m + 6 &= m + 11 \\ m + 6 &= 11 \\ m &= 5 \end{aligned}$$

(f)
$$\begin{aligned} 3(n - 2) &= 2n + 1 \\ 3n - 6 &= 2n + 1 \\ n - 6 &= 1 \\ n &= 7 \end{aligned}$$

7. (a)
$$\begin{aligned} 2(x + 5) &= 26 \\ x + 5 &= 13 \\ x &= 8 \end{aligned}$$

(b) (i)
$$\begin{aligned} 4(y - 1) &= 2y + 10 \\ 4y - 4 &= 2y + 10 \\ 2y - 4 &= 10 \\ 2y &= 14 \\ y &= 7 \end{aligned}$$

(ii) Substitute $y = 7$ into each side of the original equation.

Left-hand side $= 4(7 - 1) = 24$

Right-hand side $= 2 \times 7 + 10 = 24$

Both sides equal 24, so $y = 7$ is correct.

8. Twice the number is $2n$, and three more gives $2n + 3$.

$$\begin{aligned} 2n + 3 &= 27 \\ 2n &= 24 \\ n &= 12 \end{aligned}$$

The number is 12.

9. (a) $$\begin{aligned} 3n - 5 &= 16 \\ 3n &= 21 \\ n &= 7 \end{aligned}$$

The number is 7.

(b) $$\begin{aligned} \frac{n}{4} + 6 &= 11 \\ \frac{n}{4} &= 5 \\ n &= 20 \end{aligned}$$

The number is 20.

10. The total cost is the fixed fee plus the hourly charge.

$$\begin{aligned} 8 + 6h &= 38 \\ 6h &= 30 \\ h &= 5 \end{aligned}$$

Mia hired the bike for 5 hours.

11. The perimeter is twice the width plus twice the length.

$$\begin{aligned} 2x + 2(x + 3) &= 30 \\ 2x + 2x + 6 &= 30 \\ 4x + 6 &= 30 \\ 4x &= 24 \\ x &= 6 \end{aligned}$$

The width is 6 cm. The length is $(6 + 3)$ cm $= 9$ cm.

Check: the perimeter is $(2 \times 6 + 2 \times 9)$ cm $= 30$ cm.

12. At 2 km, the distance charge is $2 \times 2 = 4$ dollars. The booking fee is therefore $11 - 4 = 7$ dollars.

Let d be the distance travelled in kilometres. For a total cost of \$23:

$$\begin{aligned} 7 + 2d &= 23 \\ 2d &= 16 \\ d &= 8 \end{aligned}$$

The distance travelled is 8 km.

13. Substitute $C = 24$ into $C = 4t + 8$.

$$\begin{aligned} 4t + 8 &= 24 \\ 4t &= 16 \\ t &= 4 \end{aligned}$$

The hire time is 4 hours. To check on the graph, read across from $C = 24$ to the line, then down to the time axis: $t = 4$.